

Name: _____

Date: _____

3. What will be the equilibrium temperature when a 265 g block of copper at 245°C is placed in a 145 g aluminum calorimeter cup containing 825 g of water at 12.0°C? The specific heat capacity for copper is 390 J/(kg·°C), for aluminum is 900 J/(kg·°C) and for water is 4180 J/(kg·°C). Show all your work.

$$M_c = 265 \times 10^{-3} \quad T_c = 245^\circ\text{C}$$

$$\underbrace{C_{Alum}}_{\text{Alum}} \Delta T + \underbrace{C_{Water}}_{\text{Water}} \Delta T = -C_{Cu} \Delta T$$

$$M_c = 265 \times 10^{-3} \quad [4.5 \text{ marks}]$$

$$M_w = 825 \times 10^{-3}$$

$$M_{cup} = 145 \times 10^{-3}$$

$$C_c = 390$$

$$C_a = 900$$

$$C_w = 4186$$

$$C_a M_a (T_F - 12) + C_w M_w (T_F - 12) =$$

$$(T_F - 12) (C_a M_a + C_w M_w) = -C_c M_c (T_F - 245^\circ\text{C})$$

$$T_F (C_a M_a + C_w M_w + C_c M_c) = (245) C_c M_c + 12 (C_a M_a + C_w M_w)$$

$$T_F = 18.53^\circ\text{C}$$

Final answer: 18.5°C

4. Approximately how long should it take 8.2 kg of ice at 0°C to melt when it is placed in a carefully sealed Styrofoam ice chest of dimensions whose walls are 1.5 cm thick? Assume that the conductivity of Styrofoam is double that of air and that the outside temperature is 34°C. The thermal conductivity for Styrofoam is 0.023 J/s·m·°C and the latent heat of fusion for ice is 3.33×10^5 J/kg. Show all your work.

$$Q = mL_{ice}$$

Energy needed to melt

$$Q = mL_{ice}$$

$$Q = 8.2 \times 3.33 \times 10^5$$

$$Q_{ice} = 273000 \text{ J}$$

Wall is 1.5 cm thick

$$\frac{Q}{t} = \frac{A \cdot \Delta T \cdot k}{L}$$

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Total Q

$$Q = \left(2 \cdot \frac{(35 \times 25 \times 10^{-4}) (84^\circ\text{C}) \cdot 0.023}{1.5 \times 10^{-2}} + 2 \cdot \frac{(25 \times 55 \times 10^{-4}) (34^\circ\text{C}) \cdot 0.023}{1.5 \times 10^{-2}} \right) \times 0.023 \times 34$$

$$627275 +$$

$$27275 =$$

$$27275 = \left(\frac{Q}{t} \right) = (4353 \text{ J/s})$$

Final answer: 117 hours

(Correct answer!)

THINKING/INQUIRY

The drawing shows two thermometers, W and Z, whose temperatures are measured in °W and °Z. The ice and boiling points of water are also indicated. Use the data in the drawing to determine what would that temperature read on the Z scale when the temperature of a substance reads 25°W on the W scale.

Show all your work.

$$25^{\circ}\text{W} \text{ to } 20^{\circ}\text{Z}$$

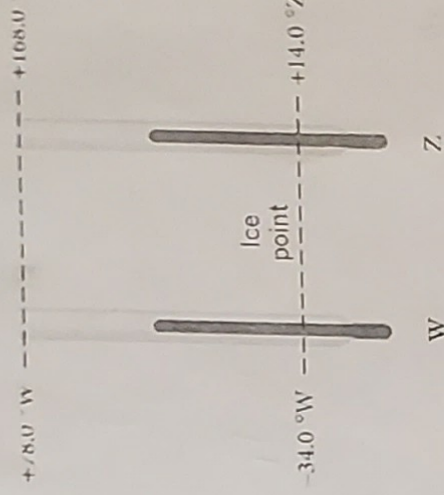
$$25^{\circ}\text{W} - (-34^{\circ}\text{W}) = 59^{\circ}\text{W} \text{ from ice point}$$

$$59^{\circ}\text{W} \cdot \frac{154^{\circ}\text{Z}}{112^{\circ}\text{W}} = 81.125^{\circ}\text{Z} \text{ from ice}$$

$$81.125^{\circ}\text{Z} + 14.0^{\circ}\text{Z} = 95.125^{\circ}\text{Z}$$

$$\text{New temp is } 95.125^{\circ}\text{Z}$$

[3 marks]



Final answer: 95.125

A brass plug is to be placed in a ring made of iron. At 15°C, the diameter of the plug is 8.755 cm and that of the inside of the ring is 8.741 cm. They must both be brought to what common temperature in order to fit? The coefficient of thermal expansion for: iron is $12 \times 10^{-6} \text{ 1/}^{\circ}\text{C}$ and for brass is $19 \times 10^{-6} \text{ 1/}^{\circ}\text{C}$.

Show all your work.

* Must be same

diameter

[3.5 marks]



$$\Delta L_B - \Delta L_{\text{Iron}} = 0$$

$$D_{\text{Brass}} + \Delta L_{\text{Brass}} = D_{\text{Iron}} + \Delta L_{\text{Iron}}$$

$$D_B + \Delta T \cdot \alpha_B \cdot D_B = D_I + \Delta T \cdot \alpha_I \cdot D_I$$

$$D_B + \Delta T \cdot \alpha_B \cdot D_B = D_I + \Delta T \cdot \alpha_I \cdot D_I$$

$$\Delta T (\alpha_B \cdot D_B - \alpha_I \cdot D_I) = D_I - D_B$$

$$\Delta T = \frac{D_I - D_B}{\alpha_B D_B - \alpha_I D_I}$$

$$\Delta T = -22.7^{\circ}\text{C}$$

$$-212.8^{\circ}\text{C}$$